THE IDEAL OF DIVISIBLE ELEMENTS IN A DERIVED COMPLETE RING IS NILPOTENT WITH AN EXPLICIT BOUND

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The counted proof span is unusually terse and ends with the bare sentence 'This finishes the proof.' without spelling out the filtration argument; that sentence is delivered verbatim and no intermediate step was inserted. The two references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The square-zero statement for the case of a principal ideal. invoked at proof line 27114 and again at proof lines 27121-27122. Serves as the base case and supplies the square-zero property of the kernel ideal.
- The derived complete modules form a Serre subcategory, so quotients by powers of an element remain derived complete. invoked at proof lines 27116-27119. Makes the inductive hypothesis applicable to A_t.
- The induction hypothesis with exponent 2^{r-1}. invoked at proof lines 27120-27121. Gives J_t^{N/2} = 0 on the quotient.

Source reference. Original lemma 27067: sources/stacks-more-algebra/pinned-original-source.tex, source lines 27067–27067. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 26068: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26068–26068. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let A be a ring derived complete with respect to an ideal I. Set $J = \bigcap I^n$. If I can be generated by r elements then $J^N = 0$ where $N = 2^r$.*

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Proof. When $r = 1$ this is Lemma 27067. Say $I = (f_1, \ldots, f_r)$ with $r > 1$. By Lemma 26068 the ring $A_t = A/f_r^t A$ is derived complete with respect to I and hence a fortiori derived complete with respect to $I_t = (f_1, \ldots, f_{r-1})A_t$. Observe that $A \to A_t$ sends J into $J_t = \bigcap I_t^n$. By induction $J_t^{N/2} = 0$ with $N = 2^r$. The ideal $\bigcap \mathrm{Ker}(A \to A_t) = \bigcap f_r^t A$ has square zero by the case $r = 1$. This finishes the proof. $\square$